TRACE-V2: Dataset Release

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1. Overview

This release provides two datasets from TRACE-V2 (Trajectory Reconstruction through Agent-based Cultural Emergence), an agent-based model developed to investigate the emergence and development of artistic careers in a simulated cultural field.

The datasets are exemplars from Simulation 1 of TRACE-V2. They provide access to the original simulation output and the processed artist-level data used in the multidimensional analysis of artistic career trajectories. The complete modelling study comprises 100 simulations; this release makes the two corresponding datasets from Simulation 1 available as representative examples rather than distributing the complete output of all 100 simulations.

The model generates career histories for 1,089 artist-agents over 260 weekly simulation steps. These histories are produced within an artificial society through a parsimonious model informed by theoretical and empirical research. The simulated agents and their career histories are computational constructs; they do not represent actual individual artists.

The two datasets serve complementary purposes:

- **Dataset 1** contains the original simulation output, with observations recorded for each artist-agent at each simulation step.
- **Dataset 2** contains the processed artist-level data, including career sequences, sequence descriptors, cluster assignments and multidimensional career configurations derived from the trajectory analyses.

2. Dataset 1: Original simulation output

Filename: simulation_data_1.csv
Format: CSV
Dimensions: 283,140 data rows × 25 columns
File size: 30.1 MB

2.1 Origin and purpose

This dataset is the original output of one complete TRACE-V2 simulation. It records the characteristics, activities and career outcomes of 1,089 artist-agents across 260 weekly simulation steps.

Each row represents one artist-agent at one simulation step. The dataset therefore contains 283,140 artist-step observations (1,089 artists × 260 steps).

The dataset provides the longitudinal information generated by the TRACE-V2 model, including changes in economic capital, symbolic capital and self-realisation, as well as artistic practice, secondary employment and exits from and returns to artistic activity.

2.2 Production

Produced by: TRACE-V2 agent-based model
Model and dataset developed by: Dr Simone Zarpelon Leao

2.3 Additional documentation

Documentation on TRACE-V2 and its theoretical and methodological foundations is available through the project repository and the associated Zenodo record: https://doi.org/10.5281/zenodo.23354544

The paper "*Sustaining an Artistic Career: A Model of Work, Time, Economic Survival, Recognition, and Self-Realisation"* is currently under journal review. A publication link will be added when available.

2.4 Data dictionary

No.	Variable	Description
1	Tick	Simulation time-step, representing weekly intervals from 0 to 259 (260 steps in total).
2	ArtistID	Unique identifier assigned to each artist-agent within the simulation.
3	EC	Economic capital held by the artist-agent at the corresponding simulation step.
4	SC	Symbolic capital held by the artist-agent at the corresponding simulation step, representing reputation and recognition within the artistic field.
5	SR	Accumulated self-realisation associated with the artist-agent's artistic practice at the corresponding simulation step.
6	State	Current participation status of the artist-agent: Active or Exited.
7	Colour	Model-assigned colour identifier used to represent the artist-agent's current status or condition in the model's visualisation.
8	InvestmentDisposition	Artist-agent disposition or propensity to allocate resources to artistic practice.
9	SelfRealisationPropensity	Artist-agent propensity to derive self-realisation from artistic practice.
10	Investment	Amount of economic resources allocated to artistic practice at the corresponding simulation step.
11	RecognitionEvent	Indicator or record of a recognition event affecting the artist-agent's symbolic capital.
12	SCGain	Change or gain in symbolic capital resulting from recognition-related processes.
13	EconomicEvent	Indicator or record of an economic event affecting the artist-agent's economic capital.
14	ECGain	Change or gain in economic capital resulting from economic activity or other modelled economic events.
15	SRGain	Gain in self-realisation associated with artistic practice at the corresponding simulation step.
16	JobSeeker	Indicator of whether the artist-agent is seeking secondary employment.
17	SecondaryJobType	Type or category of secondary employment undertaken by the artist-agent at the corresponding simulation step.
18	SecondaryJobHours	Number of hours allocated to secondary employment at the corresponding simulation step.
19	SecondaryJobHourlyWage	Hourly wage associated with the artist-agent's secondary employment.
20	SecondaryJobDuration	Duration associated with the artist-agent's secondary employment.
21	JobECGain	Economic-capital gain generated through secondary employment.
22	ArtisticHours	Number of hours allocated to artistic practice at the corresponding simulation step.
23	ExitCount	Cumulative number of exit events recorded for the artist-agent up to the corresponding simulation step.
24	ReturnCount	Cumulative number of return events recorded for the artist-agent up to the corresponding simulation step.
25	SecondaryJobCount	Cumulative number of secondary jobs recorded for the artist-agent up to the corresponding simulation step.

3. Dataset 2: Processed artist-level data and multidimensional trajectory classifications

Filename: simulation_1_all_variables_all_four_clusters.csv
Format: CSV
Dimensions: 1,089 data rows × 68 columns
File size: 4.6 MB

3.1 Origin and purpose

This dataset contains the processed analytical results for the 1,089 artist-agents in Simulation 1 of TRACE-V2.

It was produced through a series of sequence analyses designed to examine artistic career trajectories from four distinct analytical perspectives:

1. **Economic capital:** trajectories of economic-capital development.
2. **Symbolic capital:** trajectories of symbolic-capital development.
3. **Secondary employment:** trajectories of engagement with secondary employment.
4. **Exits:** trajectories relating to exits from and returns to artistic practice.

Each perspective was analysed separately through sequence analysis, optimal matching and cluster analysis. The resulting classifications are represented in the dataset by four corresponding cluster assignments.

The dataset also includes artist-level career descriptors, sequence characteristics and measures summarising the distribution and complexity of each artist's trajectory. The final variable, Multidimensional_Career, identifies the combination of the four cluster assignments for each artist-agent.

The original longitudinal histories are therefore represented here through artist-level records and derived trajectory measures, rather than through one observation per artist per simulation step.

3.2 Production

Original simulation data: TRACE-V2 agent-based model
Processed dataset and sequence analyses: Developed by Dr Simone Zarpelon Leao using R.

3.3 Additional documentation

Documentation on TRACE-V2 and its theoretical and methodological foundations is available through the project repository and the associated Zenodo record:

https://doi.org/10.5281/zenodo.23354544

The paper "*The Multidimensionality of Artistic Careers: Analytical Choices and the Construction of Trajectory Typologies*" is currently under review. A publication link will be added when available.

3.4 Data dictionary

No.	Variable	Description
1	ArtistID	Unique identifier assigned to each artist-agent within the simulation.
2	E_sequence	Sequence of economic-capital categories (E1–E5) experienced by the artist-agent across the 260 simulation steps.
3	final_Tick	Last simulation step recorded for the artist-agent. For a complete simulation, this is tick 259.
4	final_SC	Symbolic capital recorded at the artist-agent's final simulation step.
5	final_EC	Economic capital recorded at the artist-agent's final simulation step.
6	final_SR	Self-realisation recorded at the artist-agent's final simulation step.
7	final_SecondaryJobCount	Cumulative number of secondary jobs recorded for the artist-agent at the final simulation step.
8	final_ExitCount	Cumulative number of exit events recorded for the artist-agent at the final simulation step.

9	final_ReturnCount	Cumulative number of return events recorded for the artist-agent at the final simulation step.
10	final_State	Artist-agent's participation status (Active or Exited) at the final simulation step.
11	mean_ECGain	Mean economic-capital gain per simulation step over the artist-agent's recorded history.
12	mean_SRGain	Mean self-realisation gain per simulation step over the artist-agent's recorded history.
13	mean_SecondaryJobHours	Mean number of hours allocated to secondary employment per simulation step over the artist-agent's recorded history.
14	mean_ArtisticHours	Mean number of hours allocated to artistic practice per simulation step over the artist-agent's recorded history.
15	Exit_spell	Total number of simulation steps during which the artist-agent's State was Exited. This measures accumulated time outside active artistic practice; it is not the number of exit events or the number of separate exit spells.
16	sequence_length	Number of observations in the artist-agent's sequence.
17	TE_transition_count	Number of transitions between consecutive economic-capital categories in the artist-agent's sequence.
18	TE_transition_rate	Rate of transitions between consecutive economic-capital categories relative to sequence length.
19	TE_longitudinal_entropy	Measure of the diversity and distribution of economic-capital categories across the sequence. Higher values indicate greater diversity in the categories represented.
20	TE_turbulence	Measure of the complexity and variation of the economic-capital sequence.
21	proportion_E1	Proportion of the artist-agent's economic-capital sequence spent in category E1.
22	proportion_E2	Proportion of the artist-agent's economic-capital sequence spent in category E2.
23	proportion_E3	Proportion of the artist-agent's economic-capital sequence spent in category E3.
24	proportion_E4	Proportion of the artist-agent's economic-capital sequence spent in category E4.
25	proportion_E5	Proportion of the artist-agent's economic-capital sequence spent in category E5.
26	TE_ClusterNumber	Numeric identifier of the cluster assigned through economic-capital sequence clustering.
27	TE_cluster	Categorical label of the artist-agent's economic-capital trajectory cluster (TE1–TE4).
28	S_sequence	Sequence of symbolic-capital categories (S1–S5) experienced by the artist-agent across the simulation steps.
29	TS_transition_count	Number of transitions between consecutive symbolic-capital categories in the artist-agent's sequence.
30	TS_transition_rate	Rate of transitions between consecutive symbolic-capital categories relative to sequence length.
31	TS_longitudinal_entropy	Measure of the diversity and distribution of symbolic-capital categories across the sequence.
32	TS_turbulence	Measure of the complexity and variation of the symbolic-capital sequence.
33	proportion_S1	Proportion of the artist-agent's symbolic-capital sequence spent in category S1.

34	proportion_S2	Proportion of the artist-agent's symbolic-capital sequence spent in category S2.
35	proportion_S3	Proportion of the artist-agent's symbolic-capital sequence spent in category S3.
36	proportion_S4	Proportion of the artist-agent's symbolic-capital sequence spent in category S4.
37	proportion_S5	Proportion of the artist-agent's symbolic-capital sequence spent in category S5.
38	TS_ClusterNumber	Numeric identifier of the cluster assigned through symbolic-capital sequence clustering.
39	TS_cluster	Categorical label of the artist-agent's symbolic-capital trajectory cluster (TS1–TS5).
40	SJ_sequence	Sequence of secondary-employment categories (SJ1–SJ6) experienced by the artist-agent across the simulation steps.
41	TJ_transition_count	Number of transitions between consecutive secondary-employment categories in the artist-agent's sequence.
42	TJ_transition_rate	Rate of transitions between consecutive secondary-employment categories relative to sequence length.
43	TJ_longitudinal_entropy	Measure of the diversity and distribution of secondary-employment categories across the sequence.
44	TJ_turbulence	Measure of the complexity and variation of the secondary-employment sequence.
45	proportion_SJ1	Proportion of the artist-agent's secondary-employment sequence spent in category SJ1.
46	proportion_SJ2	Proportion of the artist-agent's secondary-employment sequence spent in category SJ2.
47	proportion_SJ3	Proportion of the artist-agent's secondary-employment sequence spent in category SJ3.
48	proportion_SJ4	Proportion of the artist-agent's secondary-employment sequence spent in category SJ4.
49	proportion_SJ5	Proportion of the artist-agent's secondary-employment sequence spent in category SJ5.
50	proportion_SJ6	Proportion of the artist-agent's secondary-employment sequence spent in category SJ6.
51	TJ_ClusterNumber	Numeric identifier of the cluster assigned through secondary-employment sequence clustering.
52	TJ_cluster	Categorical label of the artist-agent's secondary-employment trajectory cluster (TJ1–TJ4).
53	Exit_sequence	Sequence of exit-related categories (X0–X5) experienced by the artist-agent across the simulation steps.
54	final_Exit_category	Exit-related category assigned to the artist-agent at the final simulation step.
55	final_AccumulatedExits	Accumulated exit-related measure recorded for the artist-agent at the final simulation step.
56	TX_transition_count	Number of transitions between consecutive exit-related categories in the artist-agent's sequence.
57	TX_transition_rate	Rate of transitions between consecutive exit-related categories relative to sequence length.

58	TX_longitudinal_entropy	Measure of the diversity and distribution of exit-related categories across the sequence.
59	TX_turbulence	Measure of the complexity and variation of the exit-related sequence.
60	proportion_X0	Proportion of the artist-agent's exit-related sequence spent in category X0.
61	proportion_X1	Proportion of the artist-agent's exit-related sequence spent in category X1.
62	proportion_X2	Proportion of the artist-agent's exit-related sequence spent in category X2.
63	proportion_X3	Proportion of the artist-agent's exit-related sequence spent in category X3.
64	proportion_X4	Proportion of the artist-agent's exit-related sequence spent in category X4.
65	proportion_X5	Proportion of the artist-agent's exit-related sequence spent in category X5.
66	TX_ClusterNumber	Numeric identifier of the cluster assigned through exit-related sequence clustering.
67	TX_cluster	Categorical label of the artist-agent's exit-related trajectory cluster (TX1–TX4).
68	Multidimensional_Career	Identifier representing the combination of the artist-agent's four trajectory cluster assignments: economic capital (TE), symbolic capital (TS), secondary employment (TJ) and exits (TX).

4. Notes on interpretation and use

The two datasets should be used together where the research objective requires connecting the original simulated career histories with the subsequent trajectory analyses.

Dataset 1 provides the longitudinal simulation output at the artist-step level. Dataset 2 provides a processed artist-level representation of the same simulation, including the four separately derived trajectory classifications and their combined configuration.

The four trajectory classifications in Dataset 2 represent distinct analytical perspectives on artistic careers. They were derived separately; they should not be interpreted as a single joint classification. The Multidimensional_Career variable brings the four cluster assignments together to identify each artist-agent's combination of trajectory types.

The data describe outcomes generated within the TRACE-V2 computational model. They should therefore be interpreted in relation to the model's assumptions, rules and parameters, rather than as direct observations of actual artists or as predictions of real-world career outcomes.

5. Citation

When using these datasets, please cite the TRACE-V2 model documentation and the relevant research publication when available. The publication details and links will be updated following publication of the associated papers.

Dataset DOI and model documentation:
https://doi.org/10.5281/zenodo.23354544

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